# Yi-Ruei Liu

Research Assistant at CPLab @ NYCU, advised by Prof. Yu-Lun Liu. Master of Computer Science from UIUC. Focused on Computer Vision research, particularly in 3D vision and physically based rendering.

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## Education

**National Yang Ming Chiao Tung University**, Computer Science — Hsinchu, Taiwan — Sept 2026 – June 2030

**University of Illinois at Urbana-Champaign**, Computer Science — Illinois, US — Sept 2024 – Jan 2026

**National Yang Ming Chiao Tung University**, Computer Science — Hsinchu, Taiwan — Sept 2021 – Jan 2025

## Experience

**Academia Sinica**, Research Assistant — Taipei, Taiwan — June 2025 – Sept 2025 — 4 months
Led a research project development on synthesizing camera effects from real-world videos.

**CompPhotoLab, National Yang Ming Chiao Tung University**, Research Assistant — Hsinchu, Taiwan — May 2023 – June 2024 — 1 year 2 months
Researched and implemented algorithms for 3D shape reconstruction from single-view images under dynamic lighting.

## Publications

**Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery** — 2026
The first method to synthesize immersive, real-time navigable 3D urban scenes solely from multi-view satellite imagery.
Jie-Ying Lee, Yi-Ruei Liu, Shr-Ruei Tsai, Wei-Cheng Chang, Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, Yu-Lun Liu
skyfall-gs.jayinnn.dev (ECCV)

**On the Generation and Transformation of Light - A Heuristic Viewpoint** — Mar 1905
Applied quantum theory to light to explain the photoelectric effect using the concept of light quanta (photons).
Albert Einstein
en.wikipedia.org/wiki/Photoelectric_effect%23Einstein%27s_explanation (Annalen der Physik)

**The Foundation of General Relativity** — Mar 1916
Published the foundational theory of general relativity, making him internationally famous.
Albert Einstein
en.wikipedia.org/wiki/General_relativity (Annalen der Physik)

## Skills

**Computer Vision:** 3D Reconstruction, 3D Gaussian Splatting, Physics-based Rendering, Inverse Rendering

## Languages

**German**

**English**

## Interests

**Physics:** Quantum Mechanics, Quantum Computing, Quantum Information, Quantum Cryptography, Quantum Communication, Quantum Teleportation

## Selected Honors

- Machine Learning Certificate, Stanford University (2018)
- Quantum Computing Certificate, Stanford University (2018)
- Quantum Information Certificate, Stanford University (2018)